Bike number crunch

How Bike Derailleurs Work

Name _____ Date _____

Write one answer in each blank. Show how you traded ones and tens when you regrouped.

1. One parts tray holds 28 gear teeth cutouts and another holds 35. How many teeth cutouts are there in all?

28 + 35 = _____

Show your thinking.

2. A rider plans 72 pedal turns and has finished 38. How many pedal turns remain?

72 − 38 = _____

Show your thinking.

3. Two ride notes record 47 wheel turns and 18 more wheel turns. How many wheel turns do they record altogether?

47 + 18 = _____

Show your thinking.

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4. A box begins with 61 gear-number cards and 24 are used. How many gear-number cards are left?

61 − 24 = _____

Show your thinking.

5. One bike outing lasts 56 minutes and another lasts 28 minutes. How many minutes is that in all?

56 + 28 = _____

Show your thinking.

6. A gear maker checks 93 teeth and marks 47 as finished. How many teeth still need checking?

93 − 47 = _____

Show your thinking.

WONDER DAILY

How Bike Derailleurs Work

A loop with teeth

The chain loops around a front chainring and one back cog. When the pedals turn forward, the moving chain helps turn the rear wheel.

Cable and spring

A shifter pulls or releases a cable. The rear derailleur swings sideways and guides the moving chain toward the next back cog.

Two pulley jobs

The upper pulley guides the chain. The lower pulley and a spring take up extra chain, keeping the long loop snug.

The front cage

On bikes with several front chainrings, a metal cage nudges the moving chain sideways from one chainring to another.

Keep it moving

The chain must move forward to climb onto another cog. A shifter click alone cannot slide a still chain into place.

Easy or hard

A big back cog makes climbing easier. A small back cog turns the wheel farther per pedal turn but needs more push.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Write the path of a shift

How Bike Derailleurs Work · Curiosity day

Name _____ Date _____

Choose one clue. Say the whole idea, then draw or write it.



A sideways nudge



Two little wheels



Easy gear, hard gear

Draw one bike gear clue and add a small arrow to the part whose job you explain.

gear · chain · cog · teeth · derailleur · cable · spring · lever · shift · easy gear · hard gear

When the lever pulls the cable, … dictate what the rear derailleur does

The lower pulley keeps the chain snug by … dictate how it takes up slack

A big back cog feels easier because … dictate what changes for each pedal turn

Does your sentence name a part, tell what it does, and connect that job to the chain?

How Bike Derailleurs Work

01 Where does the chain travel?



The chain loops around a front chainring and one cog in the back cassette.

Look closer. Follow the chain around the front chainring, back to one cog in the cassette, and through the two small pulley wheels below.

A little explanation. The chain forms one long loop. It grips teeth on a front gear, called a chainring, and on one back gear, called a cog. Turning the pedals pulls the chain and turns the back wheel.

Try this. Use a finger in the air to trace the whole loop without touching the bike. Name the front chainring, back cog, and lower pulley wheels.

02 What does the rear derailleur move?



Cable pull and spring return move the rear derailleur sideways from cog to cog.

Look closer. Look below the back cogs for the small arm that lines its upper pulley with the cog holding the chain.

A little explanation. On a mechanical bike, the shifter pulls or lets go of a cable. The cable and a return spring move the rear derailleur sideways. Its upper pulley guides the moving chain from one back cog to the next.

Try this. Hold two hands apart like neighboring cogs. Move one finger sideways from one hand to the other to model the chain's short trip.

03 Why are there two small pulley wheels?



One pulley guides the chain while the other moves to take up slack.

Look closer. See the chain bend around an upper guide pulley and a lower tension pulley before it heads forward again.

A little explanation. Different cogs need slightly different amounts of chain. A spring turns the derailleur cage, and the lower pulley moves with it to take up slack. The upper pulley guides the chain toward the chosen cog.

Try this. Curve a piece of yarn around two bottle caps on a table. Move the lower cap gently and notice how the extra loop becomes longer or shorter.

04 Why do big and small back cogs feel different?



A big back cog makes hills easier; a small back cog gives more wheel turns on flat ground.

Look closer. Compare the largest back cog with the smallest, then find the teeth that catch the chain.

A little explanation. A big back cog is an easy gear for hills: the wheel turns less for each pedal turn, so the pedals need less push. A small back cog is a hard gear that can carry the bike farther per pedal turn on flat ground. With 32 teeth in front and 16 in back, the wheel turns 2 times for one pedal turn.

Try this. With an adult, place the bike upside down on a flat floor or secure it in a stand. Turn one pedal slowly forward by hand while only the adult shifts. Watch the derailleur move the chain. Count teeth on the smallest and biggest back cogs, then count wheel turns for one pedal turn in an easy gear and a hard gear. Keep fingers away from the chain, gears, and spokes whenever anything is turning. No riding is needed.